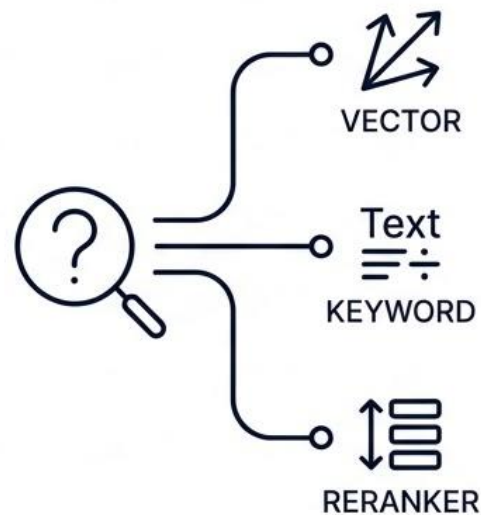


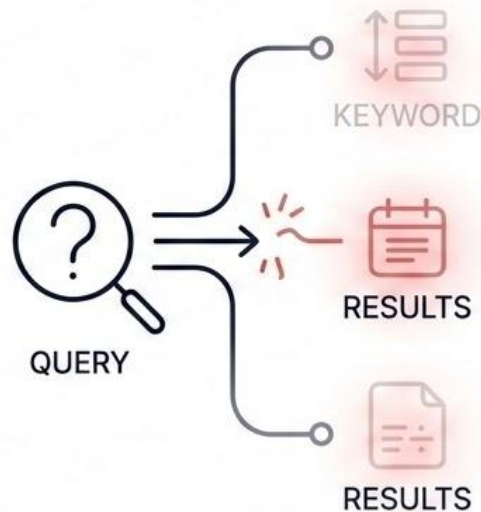
Advanced Retrieval Techniques

Beyond Simple Vector Search



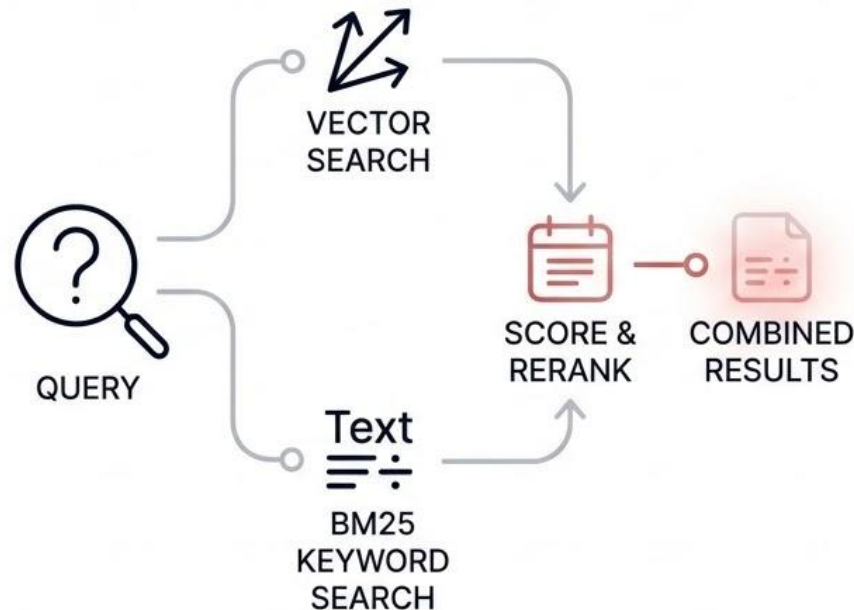
Limitations of Pure Vector Search

- Semantic similarity isn't always enough
- Struggles with exact keywords, dates, names
- Poor handling of recent or specific information
- No built-in filtering



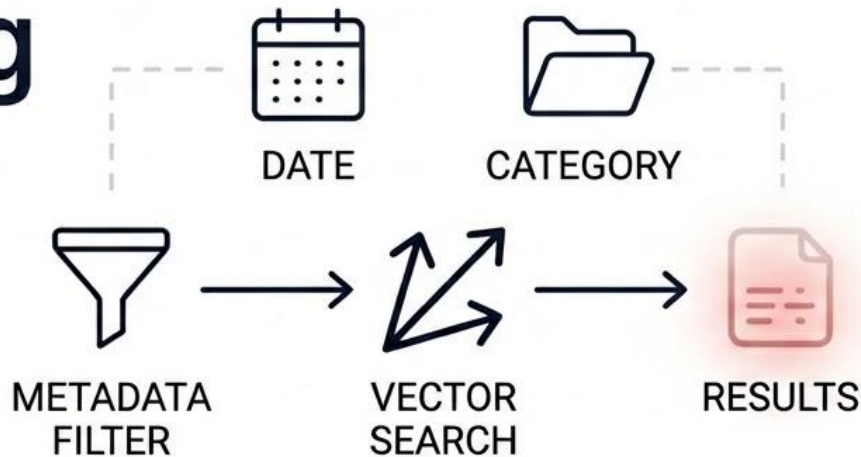
Hybrid Search: Vector + BM25

- Combines semantic (vector) + keyword (BM25) search
- Benefits: Best of both worlds - meaning + precision



Metadata Filtering

- Why it matters: Enables precise filtering before/after vector search
- Examples: date, category, source, entities



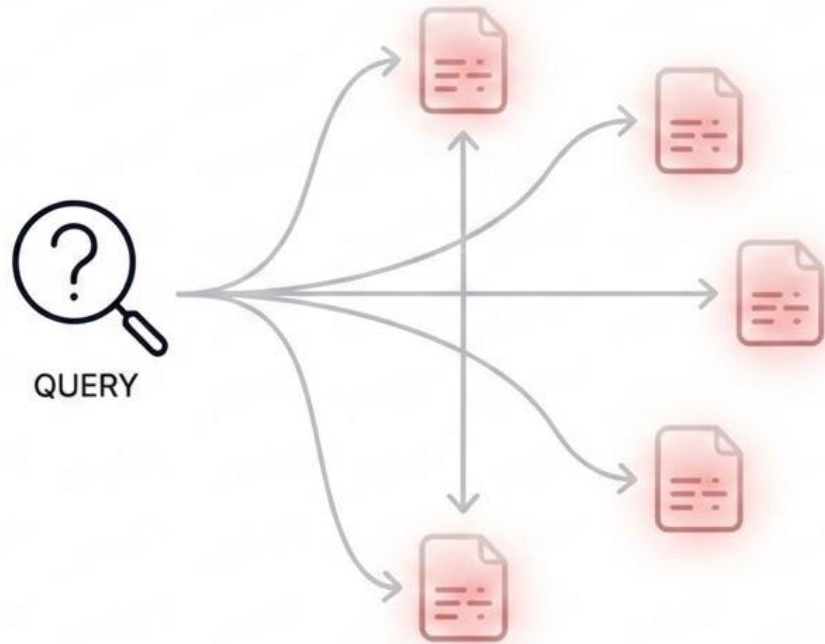
Query Expansion Techniques

Methods:

- Synonym expansion
- LLM-based query rewriting
- Multi-query generation

Benefits:

- Improves recall for complex questions

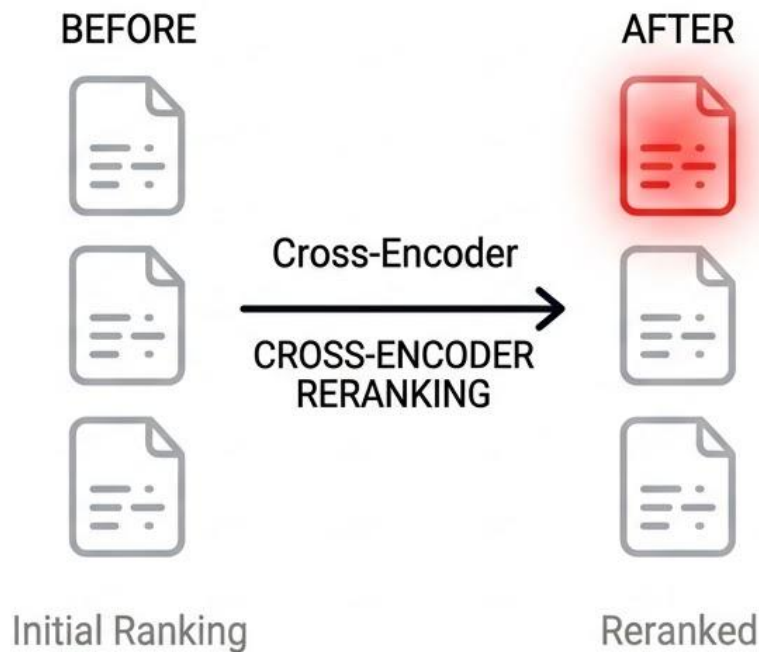


Reranking with Cross-Encoders

- Concept: Retrieve → Rerank with better model
- Popular Tools:
 - Cohere Rerank
 - BGE Reranker

BenefitsshRank

- Dramatically improves top results quality

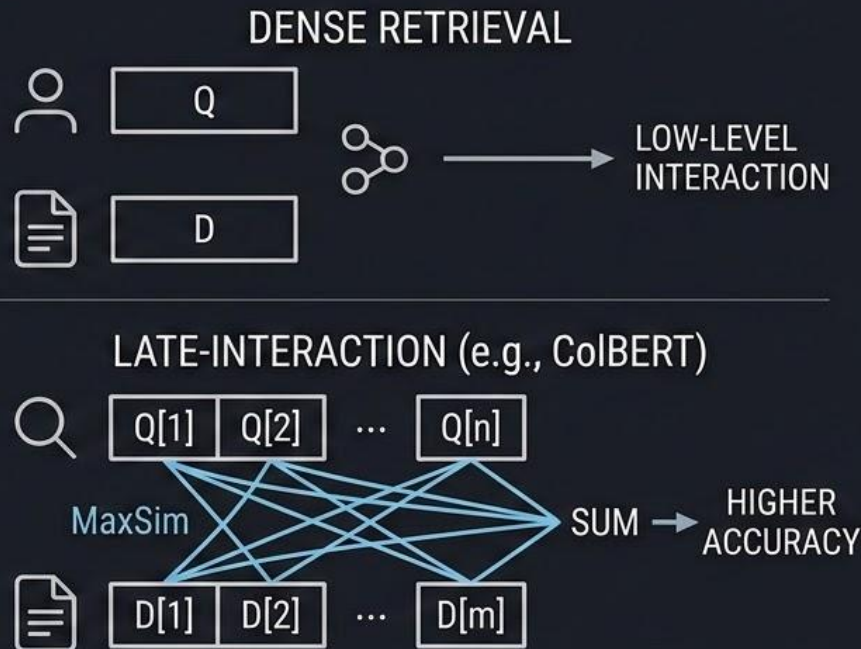


Late-Interaction Retrievers

INTRODUCTION TO:

- ColBERT (for text)
- ColPali (for multimodal/PDFs as images)

ADVANTAGE: More accurate than dense retrieval.



When Late-Interaction Beats Dense Retrieval

KEY USE CASES

- Complex, multi-term queries
- PDF/Document heavy applications
- High precision needed

PERFORMANCE FACTORS

Dense Retrieval

Late-Interaction

Handles Complex Queries

×

✓

Precision Level

Low to Mid

High

Cost vs Performance

DENSE: Lower Cost, Lower Precision | LATE:
Higher Performance, Higher Cost

×

✓

Retrieval Techniques Comparison

Technique	Speed	Accuracy	Cost	Best For
 Pure Vector	↑ High ●	↗ Mid-High ●	● Mid	General semantic search
 Hybrid (Vector+BM25)	↗ Mid-High ●	↗ High ●	● Mid-High ●	Balanced precision and recall
 Metadata Filtering	↑↑ Very High ●●	↗ High ● (for specific filters)	● Low	Structured data
 Reranking	● Mid	↑↑ Very High ●	● High	Maximizing precision on top results
 Late-Interaction (ColBERT/ColPali)	● Low-Mid	↑↑ Very High ●	● Very High	Complex queries, high-precision needs

Typical Production Retrieval Pipeline



Best Practices for Advanced Retrieval

- ✓ Always use Hybrid Search
- ✓ Add metadata filtering
- ✓ Implement reranking for top results
- ✓ Test multiple strategies
- ✓ Monitor latency vs quality tradeoff

Key Takeaways

- Pure vector search is rarely enough
- Hybrid + Reranking gives major improvements
- Late-Interaction models excel in precision
- Always measure retrieval quality

Great Retrieval = Foundation of Great RAG